

End-to-End Project Controls Analytics (Demo EMEA)



Transforming Insights into Sales

Empowering Decisions with Automated Analytics



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Role: Data & Business Analytics Engineer - Project Controls & Power BI

Organisation: GagaMuller Group

Domain: Data centres & construction

Overview

I designed and built an end-to-end **project controls analytics solution** for a demo EMEA data centre campus.

The goal was to show hyperscale data centre clients how we can connect data from **pre-construction, construction, and portfolio** levels into one digital reporting environment using **Power BI** and **Microsoft Fabric**.

What I Do

- Specialise in **end-to-end analytics solutions** for data centre and construction projects.
- Experience with **Power BI**, **Microsoft Fabric** (Lakehouse, Pipelines, Data Factory), **Python**, and **SharePoint**.
- Built complete **demo and production reporting suites** at **GagaMuller**, used by project directors, planners, and commercial teams.

My Role in This Project

- Led the **data & analytics workstream** for the Demo EMEA environment.
- Gathered requirements from internal project controls consultants and stakeholders.
- Designed the **data model and architecture** for multiple disciplines (cost, schedule, BIM, H&S, quality, procurement, portfolio).
- Built the full suite of **Power BI dashboards** and iterated based on feedback.

Tech Stack

- **Power BI** - data modelling, DAX, interactive dashboards
- **Microsoft Fabric** -Lakehouse, Pipelines, Data Factory, medallion architecture
- **Python** - ETL, Primavera P6 XER → CSV, data cleaning & transformation
- **SharePoint / Excel / BIM exports / Primavera P6** - core data sources

Pre-Construction Analytics

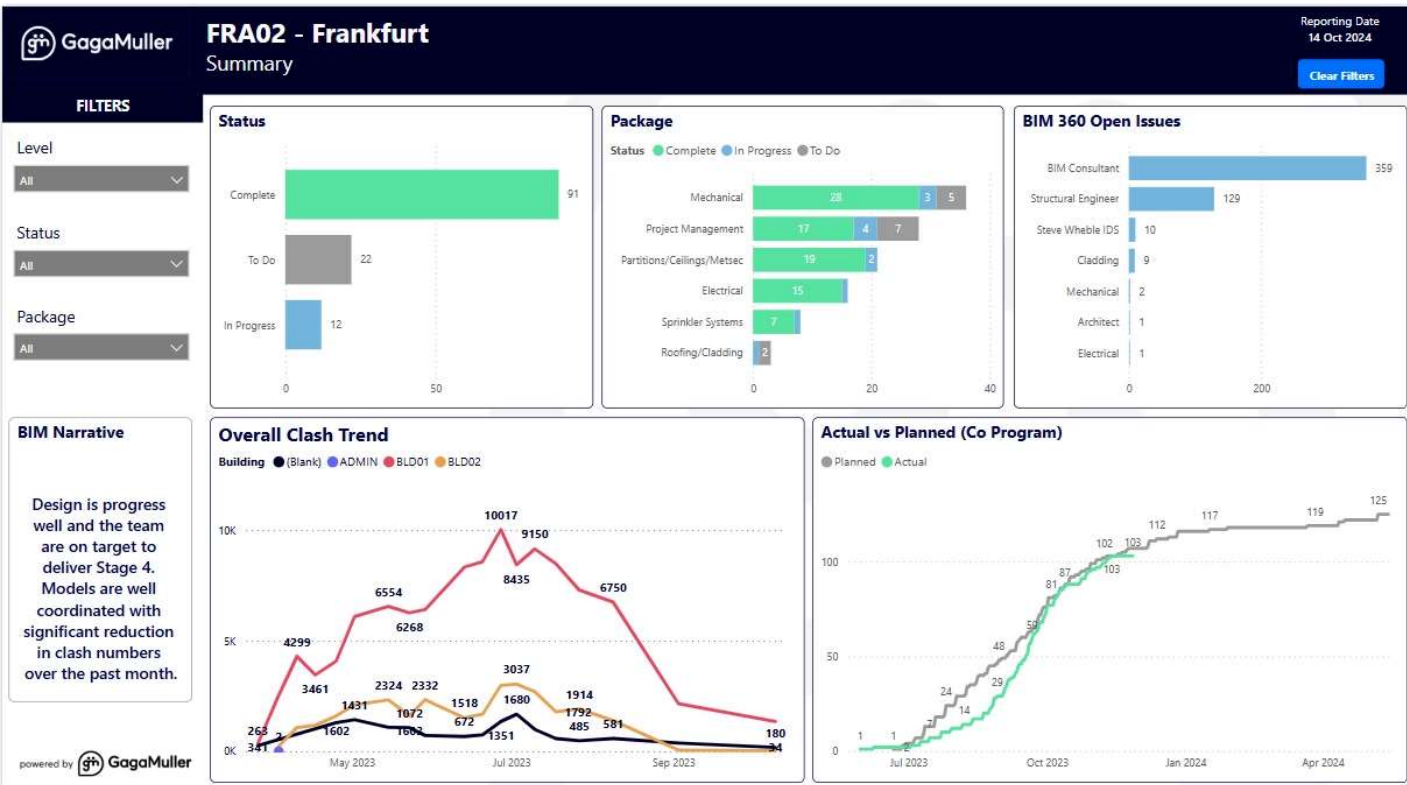
(Design, BIM & Procurement)

Objective: Give design and pre-construction teams early visibility of risk before site works start.

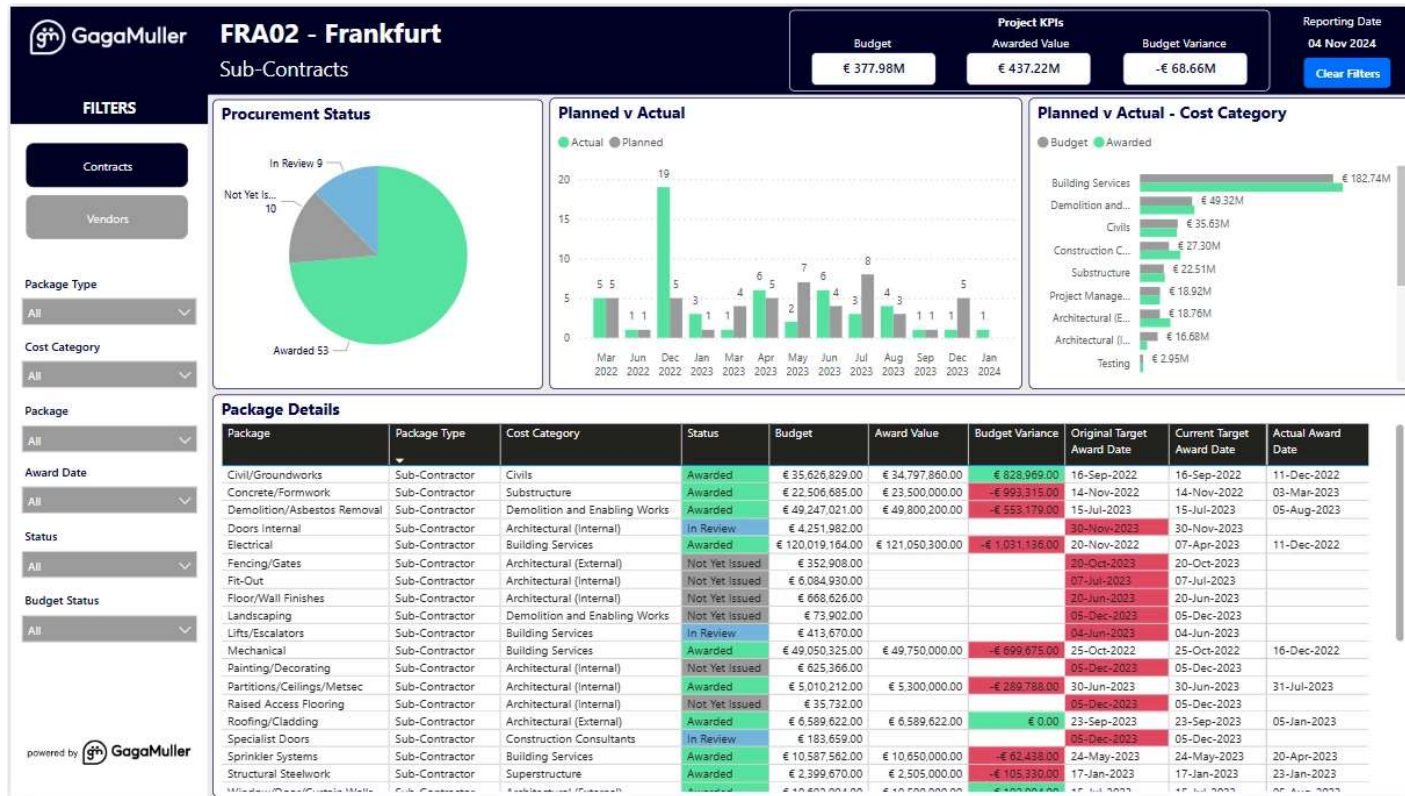
What I built

- **BIM dashboards** - open issues, clash results, and model health by package, location, and severity.
- **Design & RFI tracking** - drawing/document status, approvals vs plan, RFIs by originator and response time.
- **Procurement dashboards** - contract awards, vendor status, and lead times vs planned dates.
- **Pre-construction summary** - one page combining BIM, design, and procurement KPIs to answer: "Is this project ready to move into construction?"

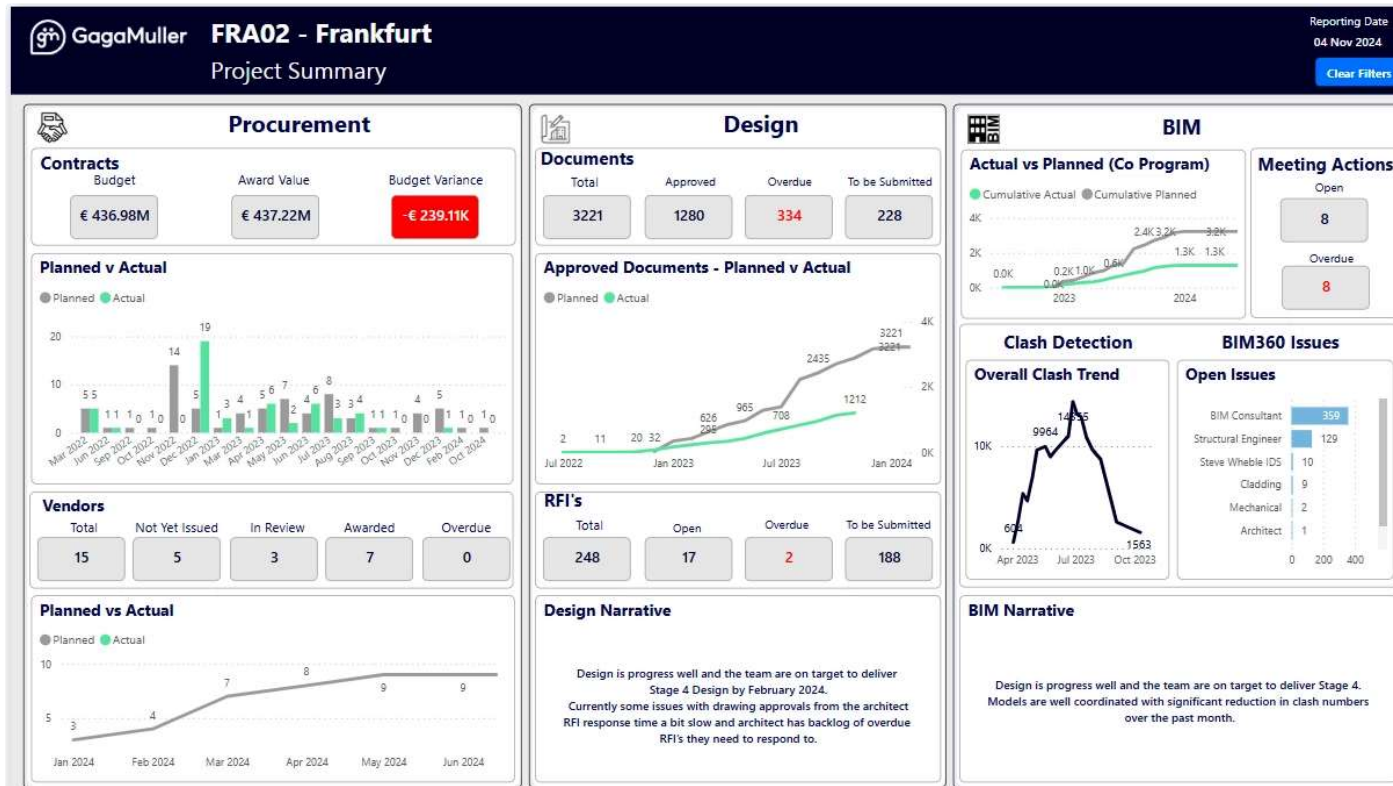
BIM summary dashboard



Design summary dashboard



Pre-construction project summary



Construction Phase Analytics

(Cost, Schedule, H&S, Quality, Commissioning)

Objective: Provide project managers and commercial teams with a live view of delivery performance.

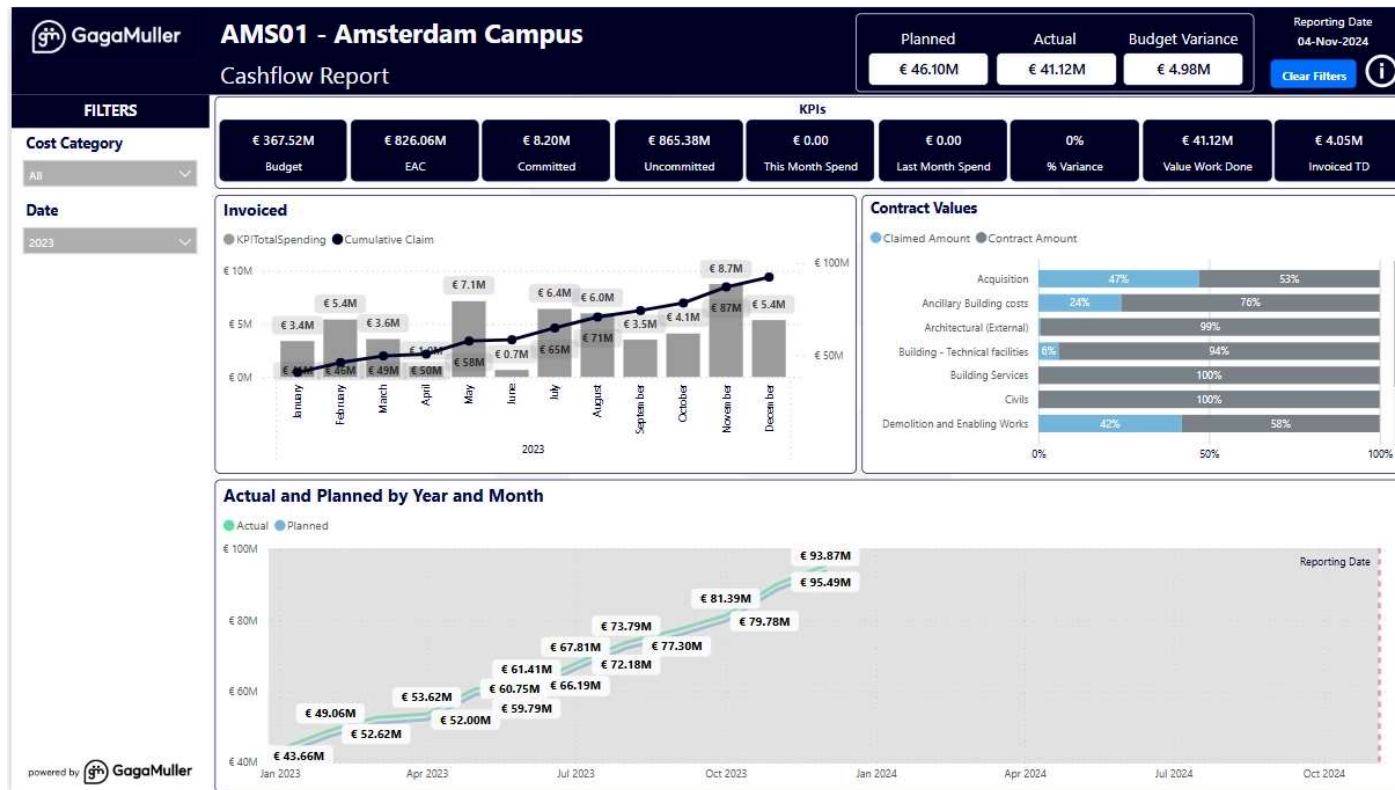
What I built

- **Cost & Cashflow** - budget vs committed vs forecast, variations, and monthly cash curve.
- **Health & Safety** - inspections, observations, and leading indicators by contractor, location, and issue type.
- **Quality** inspections, open issues, and root-cause analysis by discipline and package.
- **Schedule & Earned Value** - schedule KPIs and EVM metrics powered by data transformed from Primavera P6.
- **Commissioning & Long-lead equipment** - status of commissioning packages and critical equipment.
- **Construction project summary** - executive page combining cost, schedule, H&S, and quality KPIs for governance meetings.

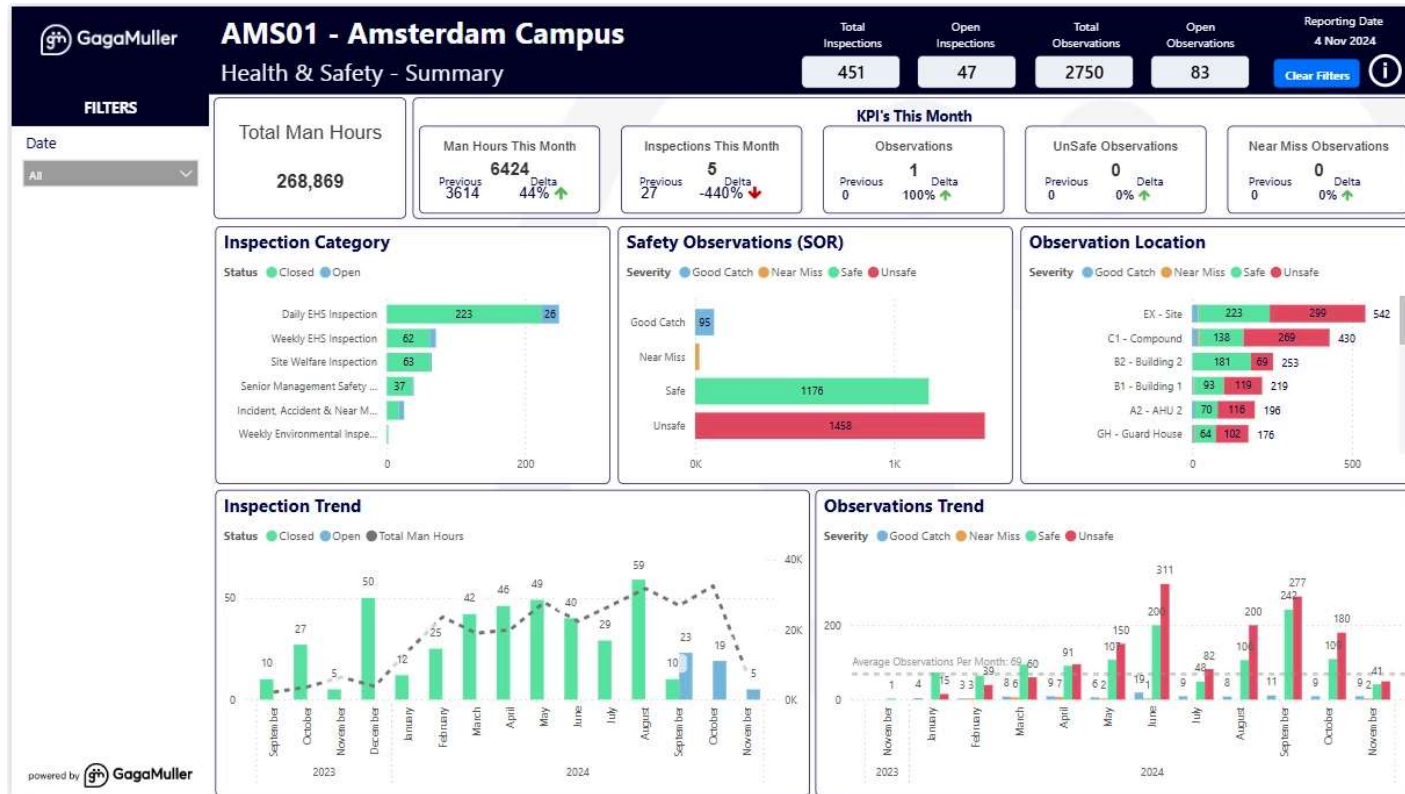
My contribution

- Integrated data from **Primavera P6 XER**, cost sheets, H&S and quality logs.
- Wrote **Python scripts** to convert XER files into structured tables for the data model.
- Ensured a **consistent UX** (shared slicers, navigation, KPI definitions) across all construction reports.

Cost / cashflow report



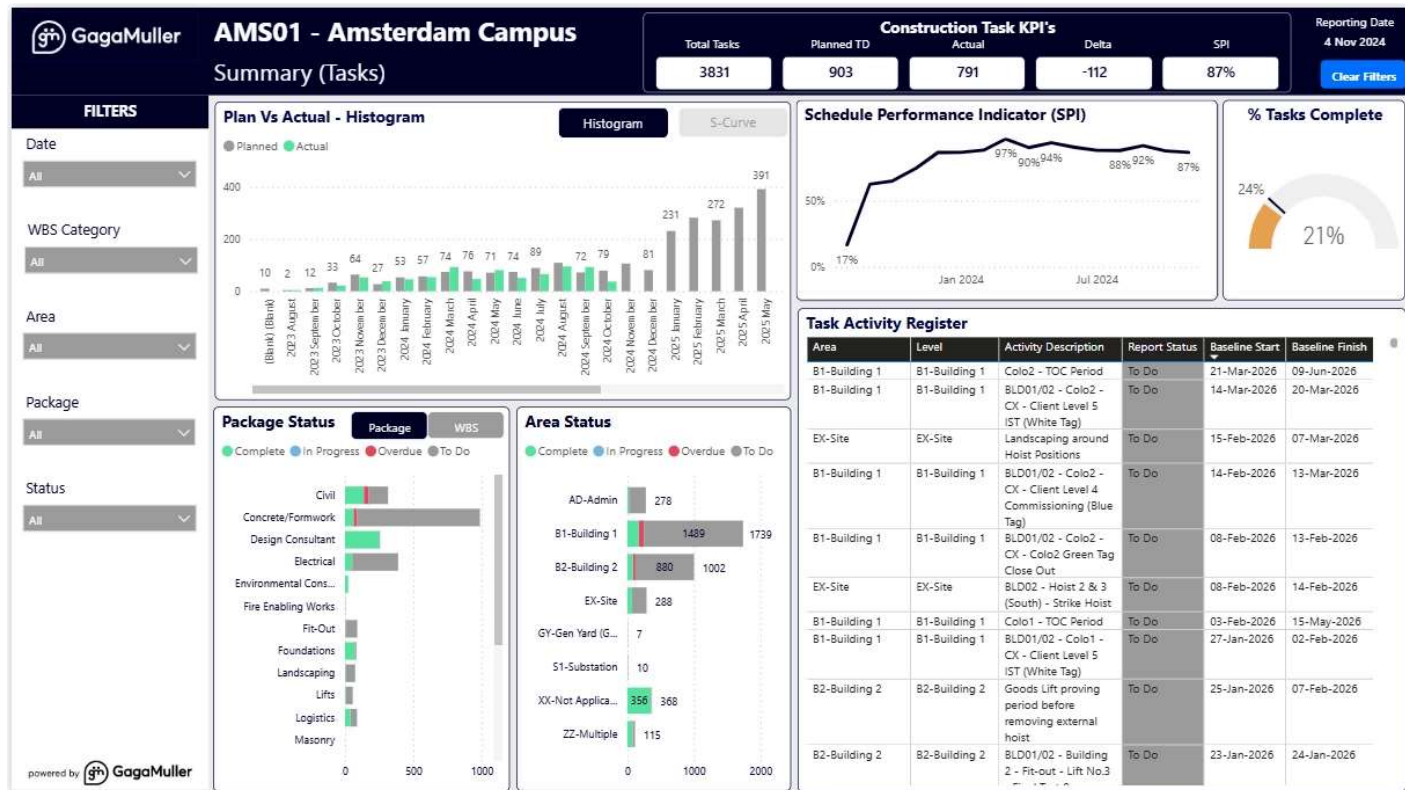
Health & Safety summary



Quality issues / inspections



Schedule / earned value dashboard



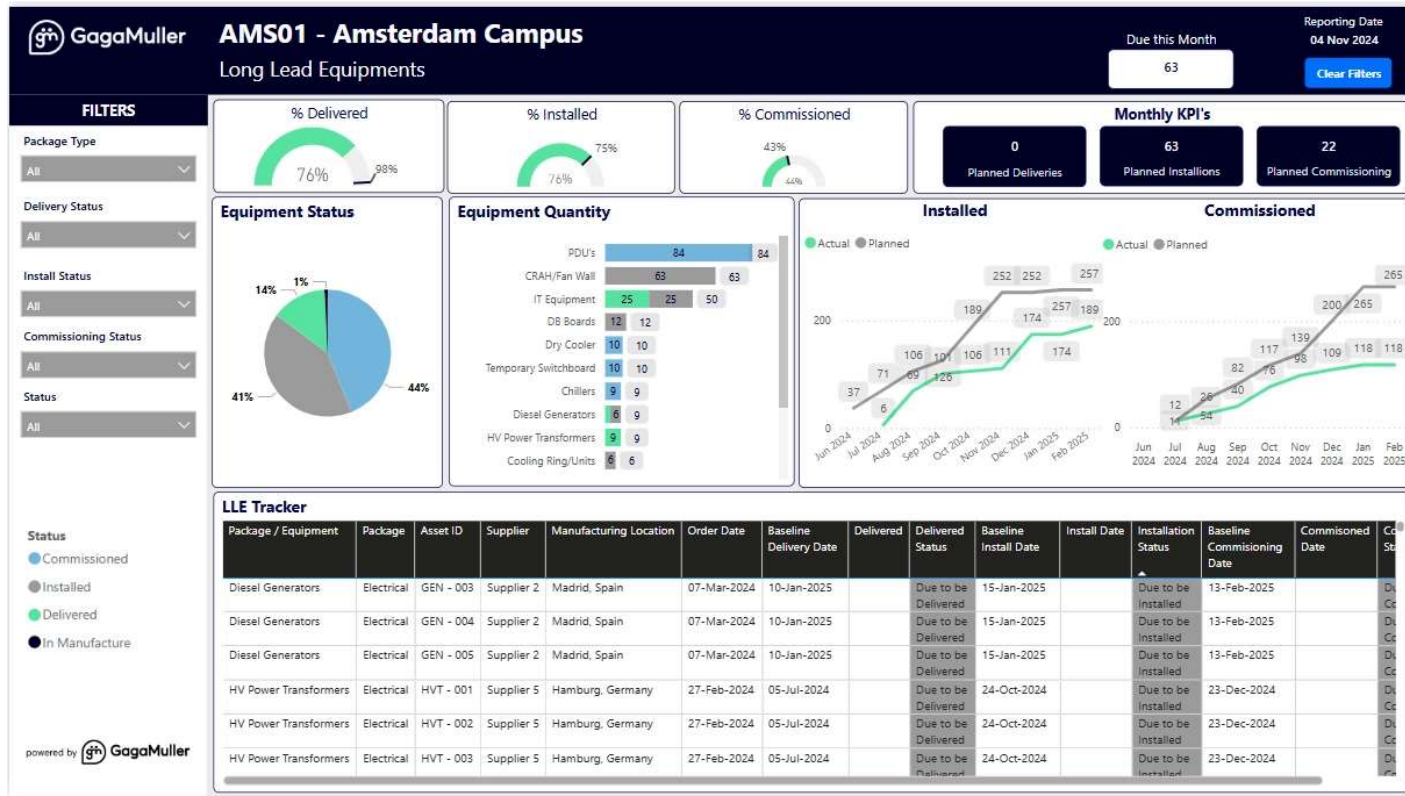
Earned Value Management



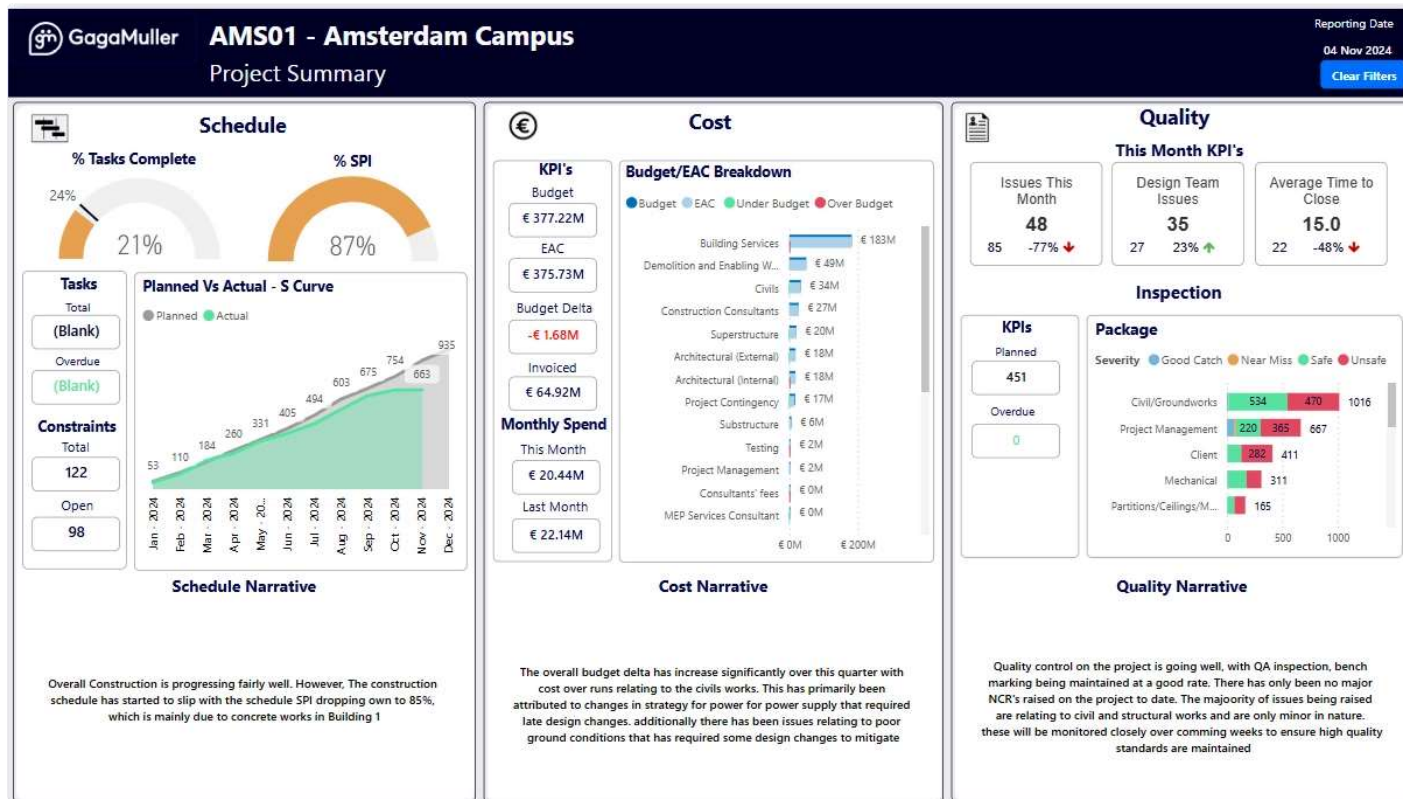
Commissioning



Long-lead equipment



Construction project summary



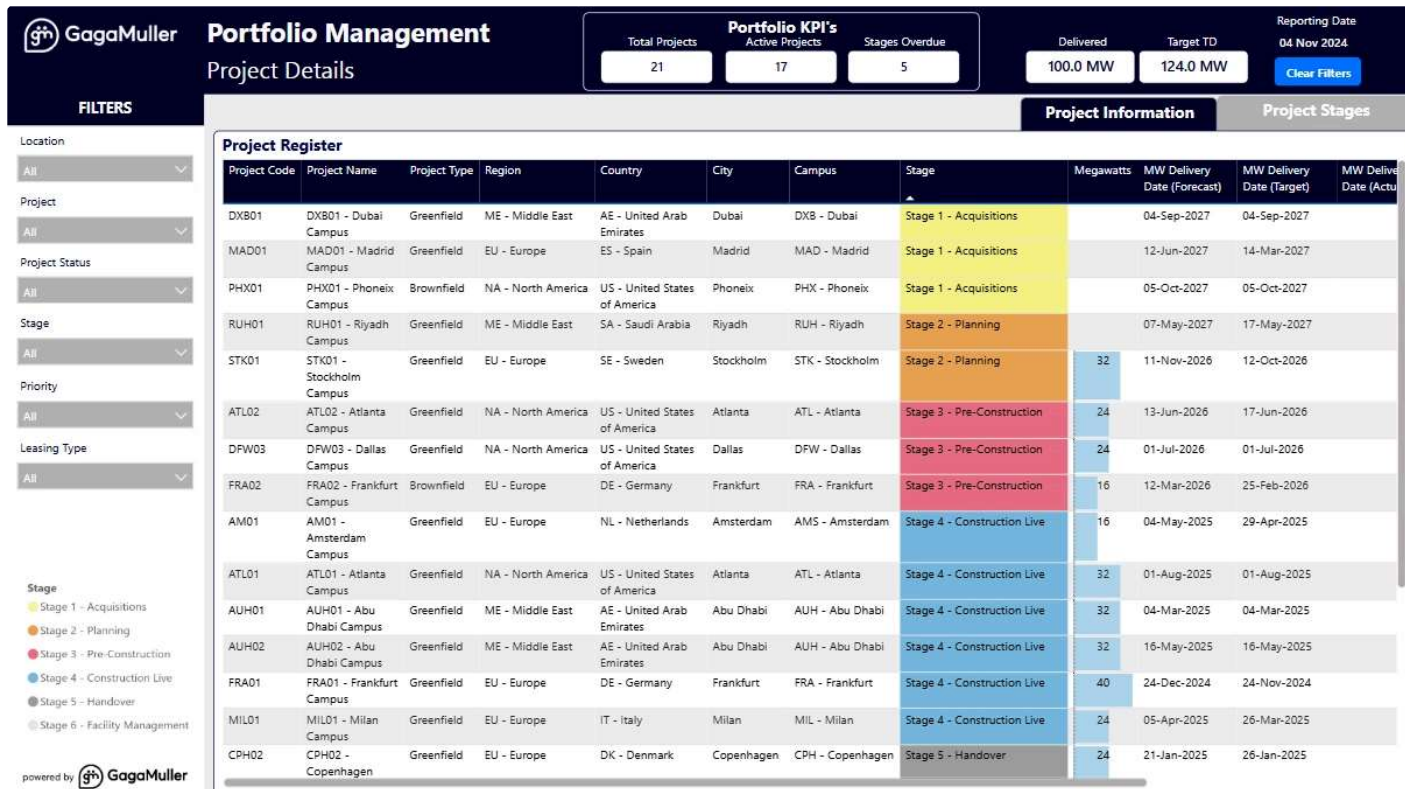
Portfolio & Programme Management

Objective: Support programme directors managing multiple data centre sites.

What I built

- **Portfolio KPIs** – schedule, stage, cost, and safety indicators across all projects, with RAG status.
- **Project list & details** – portfolio-wide table and detail view for each campus/site with milestones and current stage.
- **Stage deliverables** – open vs completed deliverables per stage and project.
- **Capacity view (where applicable)** – IT capacity vs plan across campuses.

Portfolio KPI dashboard



FILTERS

Location

All

Project

All

Project Status

All

Stage

All

Priority

All

Leasing Type

All

Stage

Stage 1 - Acquisitions

Stage 2 - Planning

Stage 3 - Pre-Construction

Stage 4 - Construction Live

Stage 5 - Handover

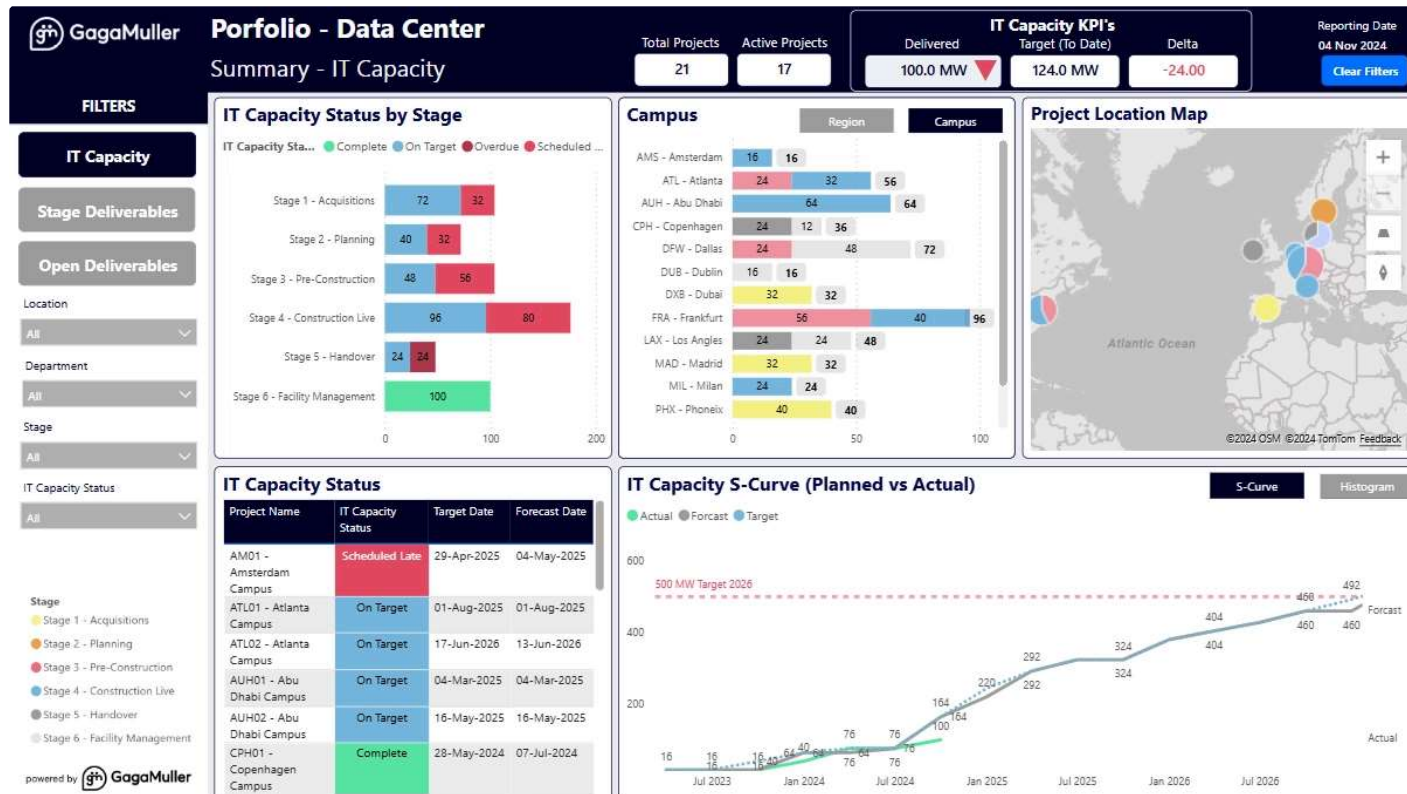
Stage 6 - Facility Management

powered by

Project Register

Project Code	Project Name	Project Type	Region	Country	City	Campus	Stage	Megawatts	MW Delivery Date (Forecast)	MW Delivery Date (Target)	MW Delivery Date (Actual)
DXB01	DXB01 - Dubai Campus	Greenfield	ME - Middle East	AE - United Arab Emirates	Dubai	DXB - Dubai	Stage 1 - Acquisitions		04-Sep-2027	04-Sep-2027	
MAD01	MAD01 - Madrid Campus	Greenfield	EU - Europe	ES - Spain	Madrid	MAD - Madrid	Stage 1 - Acquisitions		12-Jun-2027	14-Mar-2027	
PHX01	PHX01 - Phoenix Campus	Brownfield	NA - North America	US - United States of America	Phoenix	PHX - Phoenix	Stage 1 - Acquisitions		05-Oct-2027	05-Oct-2027	
RUH01	RUH01 - Riyadh Campus	Greenfield	ME - Middle East	SA - Saudi Arabia	Riyadh	RUH - Riyadh	Stage 2 - Planning		07-May-2027	17-May-2027	
STK01	STK01 - Stockholm Campus	Greenfield	EU - Europe	SE - Sweden	Stockholm	STK - Stockholm	Stage 2 - Planning	32	11-Nov-2026	12-Oct-2026	
ATL02	ATL02 - Atlanta Campus	Greenfield	NA - North America	US - United States of America	Atlanta	ATL - Atlanta	Stage 3 - Pre-Construction	24	13-Jun-2026	17-Jun-2026	
DFW03	DFW03 - Dallas Campus	Greenfield	NA - North America	US - United States of America	Dallas	DFW - Dallas	Stage 3 - Pre-Construction	24	01-Jul-2026	01-Jul-2026	
FRA02	FRA02 - Frankfurt Campus	Brownfield	EU - Europe	DE - Germany	Frankfurt	FRA - Frankfurt	Stage 3 - Pre-Construction	16	12-Mar-2026	25-Feb-2026	
AM01	AM01 - Amsterdam Campus	Greenfield	EU - Europe	NL - Netherlands	Amsterdam	AMS - Amsterdam	Stage 4 - Construction Live	16	04-May-2025	29-Apr-2025	
ATL01	ATL01 - Atlanta Campus	Greenfield	NA - North America	US - United States of America	Atlanta	ATL - Atlanta	Stage 4 - Construction Live	32	01-Aug-2025	01-Aug-2025	
AUH01	AUH01 - Abu Dhabi Campus	Greenfield	ME - Middle East	AE - United Arab Emirates	Abu Dhabi	AUH - Abu Dhabi	Stage 4 - Construction Live	32	04-Mar-2025	04-Mar-2025	
AUH02	AUH02 - Abu Dhabi Campus	Greenfield	ME - Middle East	AE - United Arab Emirates	Abu Dhabi	AUH - Abu Dhabi	Stage 4 - Construction Live	32	16-May-2025	16-May-2025	
FRA01	FRA01 - Frankfurt Campus	Greenfield	EU - Europe	DE - Germany	Frankfurt	FRA - Frankfurt	Stage 4 - Construction Live	40	24-Dec-2024	24-Nov-2024	
MIL01	MIL01 - Milan Campus	Greenfield	EU - Europe	IT - Italy	Milan	MIL - Milan	Stage 4 - Construction Live	24	05-Apr-2025	26-Mar-2025	
CPH02	CPH02 - Copenhagen	Greenfield	EU - Europe	DK - Denmark	Copenhagen	CPH - Copenhagen	Stage 5 - Handover	24	21-Jan-2025	26-Jan-2025	

Portfolio summary



Stage deliverables dashboard

FILTERS

IT Capacity

Stage Deliverables

Open Deliverables

Location

All

Department

All

Stage

All

Stage

Status Delivera... ● Blocker ● Complete ● In Progress ● To Do



Campus

Campus



Departments

Status Delivera... ● Blocker ● Complete ● In Progress ● To Do



Overdue Deliverables

Stage Deliverables

Table

References

1. *Environ Biol Fish* 2006;77:169–177.